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Research Article

Natural Products

Elevated 7-Hydroxymitragynine Levels Found in Products Misbranded as Kratom

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Abstract

Background: The leaves of *Mitragyna speciosa*, commonly called kratom, have been consumed for centuries for their energizing and analgesic properties. Kratom has gained popularity in North America, with an estimated 16 million American consumers. Recently, a new product has emerged, labeled "kratom extract" but purportedly composed of high concentrations of 7-hydroxymitragynine, an oxidation product of mitragynine, the primary alkaloid found in kratom.

Objective: 7-hydroxymitragynine, a potent mu-agonist, may pose a serious public health risk. Commercial products labeled as containing 7-hydroxymitragynine were evaluated to determine whether the constituent was present and in what concentration.

Method: The concentrations of 7-hydroxymitragynine and mitragynine were determined by a previously validated HPLC-DAD method, AOAC *Official Method*[SM] **2017.14**. The chromatographic profiles of products were also compared to authentic kratom leaf.

Results: All products were described on their label as containing "kratom extracts." The concentration of mitragynine ranged from 2.0 to 6.0 mg/g dry weight, substantially lower than typically found in kratom leaves or extracts. 7-hydroxymitragynine was determined at 22–75 mg/g dry weight, 5–28% higher than the label claims. The chromatographic profiles of all products were inconsistent with kratom leaf, having exceptional levels of 7-hydroxymitragynine, lacking the other major naturally occurring alkaloids, and exhibiting peaks not observed in leaf.

Conclusions: The products were verified to contain very high levels of 7-hydroxymitragynine, achievable only by synthetic means. These products differ significantly from authentic native kratom leaf and are not kratom extracts, as labeled, but rather synthetic derivatives that constitute unapproved new drugs being unlawfully marketed. There is an urgent need to differentiate kratom, which has been safely consumed for centuries, from 7-hydroxymitragynine, which may pose a significant risk to public health.

Highlights: Traditional kratom leaf is distinguished from products found to contain high concentrations of 7-hydroxymitragynine, only achievable by synthetic means. These unapproved new drugs contained other unidentified constituents, also not present in the native kratom leaf. The presence of concentrated 7-hydroxymitragynine products in the market masquerading as kratom is misleading, unlawful, and potentially dangerous to the public.

Kratom is the common name attributed to the deciduous tree *Mitragyna speciosa* (Korth.) Havil., a tropical tree that is part of the Rubiaceae family, sharing a lineage with coffee and indigenous to Southeast Asia. Historically, kratom leaves were chewed or brewed to mitigate fatigue and alleviate pain, particularly among manual laborers in the hot, tropical environments of countries such as Thailand and Malaysia (1–3). More recently, kratom has gained popularity in North America, with an estimated 16 million Americans reportedly consuming kratom for a mild energy boost, to improve mood, and to alleviate stress and minor pain (2, 4–7).

Kratom leaf contains a broad spectrum of plant secondary metabolites including alkaloids, flavonoids, polyphenolic compounds, triterpenoids, triterpenoid saponins, monoterpenes, and secoiroids (2). The alkaloids are recognized as the primary active compounds, with leaves containing several alkaloids representing 0.5–3% of the total weight (8–10). The primary alkaloid is mitragynine, reported to comprise 51–82% of the total alkaloid composition weight (8–12) and on a percent weight basis ranging from <1.0% but usually not >2.5% (10, 13). The other pharmacologically active alkaloids typically found within the leaf are paynantheine, speciogynine, and speciociliatine, accounting for most of the remaining alkaloid composition.

Mitragynine is metabolized in the human body to some degree into 7-hydroxymitragynine (7-HMG), an active human metabolite (14). Although initially thought to be present in native kratom leaf material, it is now recognized that 7-HMG is generated during drying and processing as an oxidative byproduct of mitragynine, observed only in post-harvest kratom leaf at trace levels (0.003–0.04%, w/w) (15–17). The pharmacology of 7-HMG drastically differs from mitragynine and the other natural kratom alkaloids. Unlike mitragynine, which acts on multiple neurotransmitter systems in the human body, 7-HMG exerts effects similar to

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conventional opiates, with studies showing it to have a binding affinity that is up to 22 times higher than morphine at the μ-opioid receptor (18–22). A recent study has reported 7-HMG induced significant respiratory depression comparable to morphine, which was reversed by naloxone, and further they found mitragynine increased respiratory frequency and is unaffected by naloxone (23). These findings highlight the critical differences between the naturally occurring kratom alkaloid mitragynine and the semisynthetic product 7-HMG.

As kratom products have surged in popularity within the United States, a new class of products has emerged that claim exceptionally high concentrations of 7-HMG, while still being labeled as kratom or "kratom extract" (11, 24). A recent survey found that 178 brands of products sold online were labeled as containing 7-HMG with claims of potentiating the product's analgesic or euphoric effects and positioning these products as "enhanced" versions of traditional kratom (25). Given that only trace levels of 7-HMG are reported in native kratom leaf, these products would have to be synthetically created in order to achieve the levels of 7-HMG claimed (11, 24). To date, no published study has verified the veracity of 7-HMG product claims, despite the potential for carrying a substantially higher risk profile than kratom. Moreover, the mislabeling of these products as kratom is misleading to consumers, who may believe they are ingesting a botanical product with a long history of safe use.

Herein we report the analysis of a selection of readily available so-called kratom products in the U.S. market that claim elevated levels of 7-HMG. Using a previously validated official AOAC First Action *Official Method*^{SM} high-performance liquid chromatography (HPLC) method (26, 27), we aim to quantify mitragynine and 7-HMG content, to allow for an accurate comparison with traditional kratom leaf standards. The described method met all AOAC standard method performance requirements for determining mitragynine and 7-HMG in kratom finished products with RSD values of ≤1% for both analytes, and recoveries between 94 and 99% (26).

Our findings are discussed in the context of the potential risks of concentrated 7-HMG products and their legal and health ramifications. In doing so, we hope to contribute to the growing discourse on kratom's safety profile and encourage regulatory interventions that differentiate kratom from synthetic unapproved drugs and protect consumers from potentially misleading product claims.

Experimental

Test Materials

Tea-cut kratom leaf was obtained from a supplier in Indonesia. Commercial products marketed as having elevated levels of 7-HMG were purchased from U.S. retailers. A total of five product brands were obtained. For Brand 1, three separate packages with the same lot number were obtained from the same vendor, and for Brand 2 two packages with the same lot number were obtained from the same vendor. All products were stored in the original packaging at room temperature until analysis. A summary of the products with their label claim is shown in Table 1.

Reference Standards

Certified reference standards for mitragynine ($C_{23}H_{30}N_2O_4$), FW of 398.495 g/mol CAS 4098-40-2, and 7-hydroxymitragynine ($C_{23}H_{30}N_2O_5$), FW of 414.49 g/mol CAS 174418–82-7, were obtained from Chromadex Standards (Longmont, CO, USA). The purity of the standards was confirmed using quantitative nuclear magnetic resonance (qNMR).

Reagents

HPLC grade methanol, acetonitrile, formic acid, and ammonia hydroxide and ACS grade acetic acid were purchased from VWR (Mississauga, ON, Canada). LC-MS grade ammonium bicarbonate was purchased from MilliporeSigma Canada Ltd (Oakville, ON, Canada). HPLC water was obtained from a Nanopure filtration unit.

Solutions

The following solutions were prepared before analysis.

(a) *Reference standard and liquid sample diluent.*—100% methanol.
(b) *Extraction solvent.*—0.5M acetic acid in 70% methanol.
(c) *HPLC Mobile phase A.*—5.0 mM ammonium bicarbonate buffer, pH 9.50.
(d) *HPLC Mobile phase B.*—Acetonitrile.

Sample Preparation

Analysis was performed using a previously validated method, AOAC *Official Method of Analysis* **2017.14**, for the determination of mitragynine and 7-HMG in raw materials and finished products (27). Sample preparation and working standard concentrations were modified to account for the higher concentrations of 7-HMG expected to be found in the commercial products.

For kratom leaf tea-cut material, 100.0 ± 5.0 mg was weighed into a 50 mL polypropylene centrifuge tube, and 10 mL of extraction solvent was added. The tube was capped and mixed using a vortex mixer for 30 seconds and then shaken on a wrist action shaker for 30 minutes. Approximately 5 mL of the solution was then filtered through a 0.45 µm PTFE filter into a test tube. Approximately 1.5 mL of the filtered solution was transferred into an HPLC vial for analysis.

For commercial product samples, a single tablet was removed from the product packaging and weighed to determine the product tablet weight. The tablet was then crushed in a mortar and pestle until a fine powder was produced. The powder was transferred to a preweighed 50 mL polypropylene centrifuge tube. The amount of powder transferred into the tube was determined and

Table 1. Summary of products analyzed and their label claim

Product	Label claim	Tablet weight, mg
Brand 1 Package 1	15 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 7.5 mg of 7-HMG	632.1
Brand 1 Package 2	15 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 7.5 mg of 7-HMG	651.7
Brand 1 Package 3	15 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 7.5 mg of 7-HMG	632.4
Brand 2 Package 1	18 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 9 mg of 7-HMG	350.6
Brand 2 Package 2	18 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 9 mg of 7-HMG	350.1
Brand 3	<14 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering <7 mg of 7-HMG	754.5
Brand 4	18 mg of 7-HMG per tablet claimed on front of package 15 mg of 7-HMG per tablet claimed on back of package	257.0
Brand 5	18 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 9 mg of 7-HMG	655.5

recorded, and 20 mL of the extraction solution was added to the tube. The tube was capped and mixed using a vortex mixer for 30 seconds and then shaken on a wrist action shaker for 30 minutes. Approximately 5 mL of the solution was then filtered through a 0.45 μm PTFE filter into a test tube. The solution was then subjected to a 1/10 dilution by adding 1 mL of the filtered solution to 9 mL of extraction solution. The diluted solution was mixed by shaking. Approximately 1.5 mL of the diluted solution was transferred to an HPLC vial for analysis.

Standard Preparation

Stock solutions were prepared by weighing out 5 mg of each chemical standard into separate 10 mL volumetric flasks. The flasks were diluted to volume with extraction solvent. Working standard solutions were prepared by serial dilution with extraction solvent as the diluent. For mitragynine, five working standard solutions ranging from 1 to 50 mg/L were prepared. For 7-HMG, five working standard solutions ranging from 1 to 150 mg/L were prepared. All working standards were analyzed by HPLC to serve as external calibration curves.

HPLC Analysis

All samples and standards were analyzed in accordance with the HPLC conditions described in AOAC **2017.14**, the previously validated method. In brief, separation was achieved on a Kinetex® EVO C18 100 Å, 150 × 4.6 mm, 5 μm particle size column (Phenomenex, Torrance, CA) using an Agilent 1200 Series Liquid Chromatograph equipped with binary pump and degasser, temperature-controlled column compartment, autosampler, flow cell, and diode-array detector (Agilent, Mississauga, ON, Canada). The column temperature was set at 25°C, the injection volume was 5 μL, and the flow rate was 1.5 mL/min. The instrument was set for a gradient binary solvent system, as described in Table 2. The detector was set to monitor at 226 nm (4 nm bandwidth).

Quantification of Alkaloids in Samples

The mitragynine and 7-HMG content for each of the samples was determined using the external calibration curves prepared for each of the analytes. The concentrations for each alkaloid in the tablet samples were first quantified in mg/g using the following equation:

$$\frac{P_0 - b_0}{m_0} \times \frac{V}{W} \times \frac{D}{1000}$$

P_0 = peak area of target analyte in sample chromatogram, b_0 = y-intercept of calibration curve for the alkaloid, m_0 = slope of calibration curve for the alkaloid, V = volume of test solution in mL, W = weight of sample in g, and D = dilution factor.

The amount of each alkaloid in each sample per serving size was then determined by multiplying the result from the equation and the weight of one serving as per the product label.

Table 2. HPLC gradient program

Time, min	% Solvent A	% Solvent B
0.0	70	30
17.0	35	65
17.9	35	65
18.0	70	30

Results and Discussion

Employing OMA AOAC **2017.14** for the determination of mitragynine and 7-HMG within kratom leaf materials and the products labeled as "kratom extracts" but purported to have high levels of 7-HMG ensured precision and accuracy in the analysis (26, 27). The mitragynine and 7-HMG content found in each product is presented in Table 3. The results are expressed as mg of alkaloid per serving based on label directions for serving size and mg of 7-HMG per tablet. The most abundant alkaloid in the kratom leaf material is mitragynine at 11.8 mg/g or 1.18% (w/w), which is within the reported levels typically found in leaf (10, 13). 7-HMG was not detected in leaf material, with a detection limit of <0.004% (w/w) for the analytical method (26). In contrast, mitragynine was not the most abundant alkaloid in the products analyzed in this study, with all but one product containing <0.6% (w/w). The most abundant alkaloid detected in all products tested was 7-HMG, with concentrations ranging from 22 to 75 mg/g. These concentrations were 5 to 28% higher than the levels claimed on the labels of these products. One product, Brand 4, tested had differing levels of 7-HMG listed on the front and back labels with neither value reflecting the actual level of 7-HMG determined. These findings show that consumers cannot rely on label statements being made on these products and would ingest levels of 7-hydroxymitragynine significantly higher than expected.

As discussed above, 7-HMG is a human metabolite of mitragynine and is found occasionally at trace levels in kratom leaf postharvest, as a byproduct of oxidation. To achieve levels of 7-HMG as found in these commercial products would require the addition of exogenous 7-HMG or chemical conversion of mitragynine to the oxidation product. Given that mitragynine in kratom leaf is typically <1.0% and usually not exceeding 2.5% on a dry weight basis and the determined levels of 7-HMG in these products are significantly greater, it is likely that highly concentrated extracts or an isolate of kratom alkaloids is used as a substrate for the synthetic transformation. Regardless, any processes used to elevate 7-HMG content would essentially create a new synthetic product that would be so different from natural kratom leaves that it could no longer be considered "kratom" in any meaningful sense of the term.

Differences in these 7-HMG products from typical kratom leaf and leaf extracts are also demonstrated through their chromatographic profiles. Figure 1 shows chromatograms obtained from authentic kratom leaf and one of the 7-HMG products analyzed using the described HPLC method. The kratom leaf material shows a typical kratom leaf profile with mitragynine as the most abundant alkaloid and the presence of the other major kratom alkaloids, namely speciogynine, panynantheine, and speciociliatine. The absence of a peak corresponding to 7-HMG is also typical of properly harvested and stored kratom leaf material.

In contrast, the most prominent peak in the 7-HMG product chromatogram is 7-HMG, with a much smaller mitragynine peak. The other common kratom alkaloids are conspicuously absent. More alarming is the presence of peaks not observed in the native leaf material, indicating the presence of compounds not naturally occurring in kratom that are likely byproducts of the process undertaken to synthetically convert mitragynine to 7-HMG. Consumers of these products may also be exposed to potentially toxic compounds with completely unknown risk profiles.

Another significant issue observed in this study was the lack of consistency both across the brands and also within products. Figure 2 shows a chromatographic overlay of all products analyzed and shows significant variance among the different

Table 3. Mitragynine and 7-HMG content found in commercial products

Product ID	Tablet weight, mg	Mitragynine content, mg/g	Mitragynine per serving, mg	7-HMG content, mg/g	7-HMG per serving, mg	7-HMG Label claim, mg/tablet	7-HMG content, mg/tablet	7-HMG % label claim
Brand 1–1	632.1	4.0	1.2	28.0	8.9	15	17.7	118%
Brand 1–2	651.7	2.7	0.8	29.0	9.4		18.9	125%
Brand 1–3	632.4	5.6	1.8	25.3	8.0		16.0	107%
Brand 2–1	350.6	2.9	0.5	53.9	9.5	18	18.9	105%
Brand 2–2	350.1	3.2	0.5	55.6	9.8		19.5	108%
Brand 3	754.5	2.1	0.8	22.4	8.5	<14	16.9	121%
Brand 4	257.0	14.6	3.8	74.7	19.2	18 (front of package) 15 (back of package)	19.2	107% (front) 128% (back)
Brand 5	655.5	2.1	0.7	29.9	9.8	18	19.6	109%

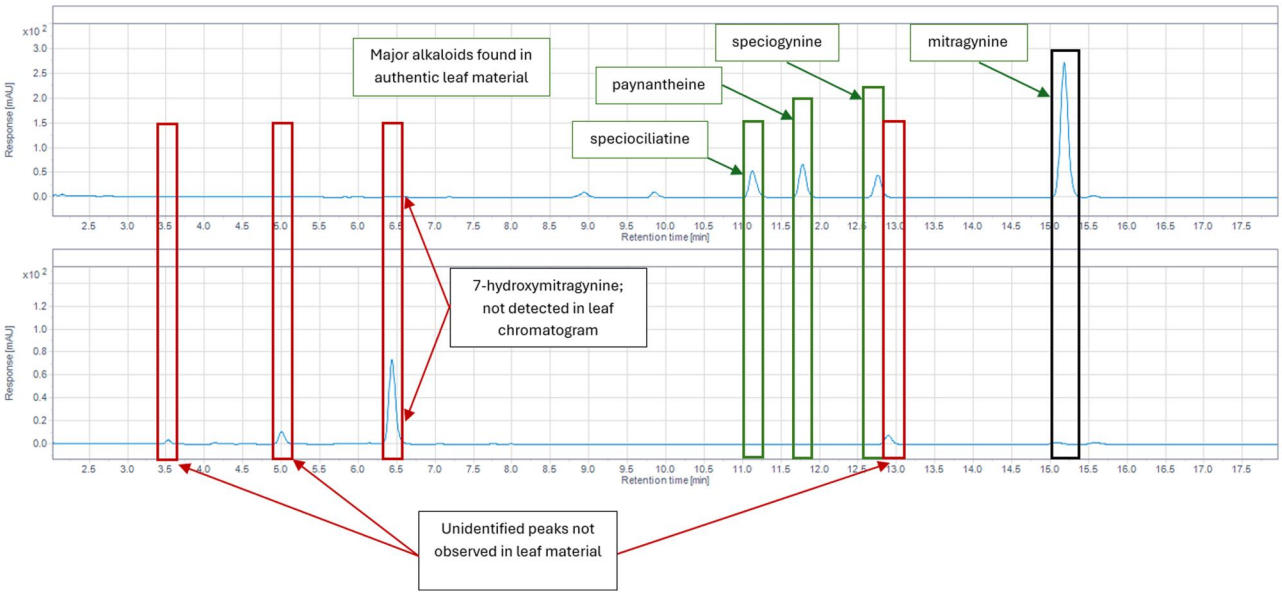


Figure 1. HPLC chromatograms of a leaf material (top) and 7-HMG product (bottom). Prominent peaks in each chromatogram are labeled.

products. All chromatograms exhibit 7-HMG as the most prominent peak; however, the presence and abundance of other peaks, both known and unknown, differ despite all products being labeled as "kratom leaf extracts." The observed differences among the products could be related to either the varying substrates used as the synthetic starting material or inconsistencies in the process itself leading to variable synthetic side- and end-products. Figure 3 shows the chromatograms obtained from analysis of the three packages of Brand 1 purchased together and shows a disturbing lack of within-product consistency. This is also evidenced in Table 3, which shows the tablets in Brand 1 contained 16.0–18.9 mg 7-HMG and 1.6–3.6 mg mitragynine per tablet. This demonstrates a lack of consistency in manufacturing and a lack of compliance with good manufacturing practices, as required by law for the safety of consumers.

Marketing these products as "kratom" is both false and misleading. Consumers unfamiliar with kratom or those relying on the package labels may erroneously purchase the products believing they are receiving a traditional kratom product, which has been associated with a long history of use in Southeast Asia (28, 29). There are an estimated 3–16 million users in the United States (30–32) and the available toxicology, clinical, and epidemiolocal studies indicate traditional kratom use has a low risk profile with respect to acute and chronic use (9, 33–37). However, this low-risk safety profile is attributed to kratom leaf and fluidized kratom leaf extracts, such as teas and decoctions, and cannot be extended to products that fall outside that definition. Consumers of 7-HMG products could be unknowingly exposing themselves to unapproved drug products with potentially higher abuse liability and unknown risks.

As discussed in the introduction, the pharmacology of 7-HMG differs from mitragynine in that it predominantly acts on opioid receptors, rather than multiple systems in the human body (18–20, 22). When available in minimal amounts and balanced with the full spectrum of alkaloids and other plant secondary metabolites, as found in traditional kratom leaf preparations, the risk of 7-HMG has never been an apparent issue. Based on ED_{50} values and minimum effective doses observed in several studies, exposure to 7-HMG for a typical kratom serving is expected to be below effect levels (9). These 7-HMG products, however, contain high concentrations of 7-HMG, do not contain the full spectrum of kratom alkaloids found within kratom leaf, and contain unknown synthetic byproducts for which there is no established safety profile. The bottom line is that these are synthetic products whose

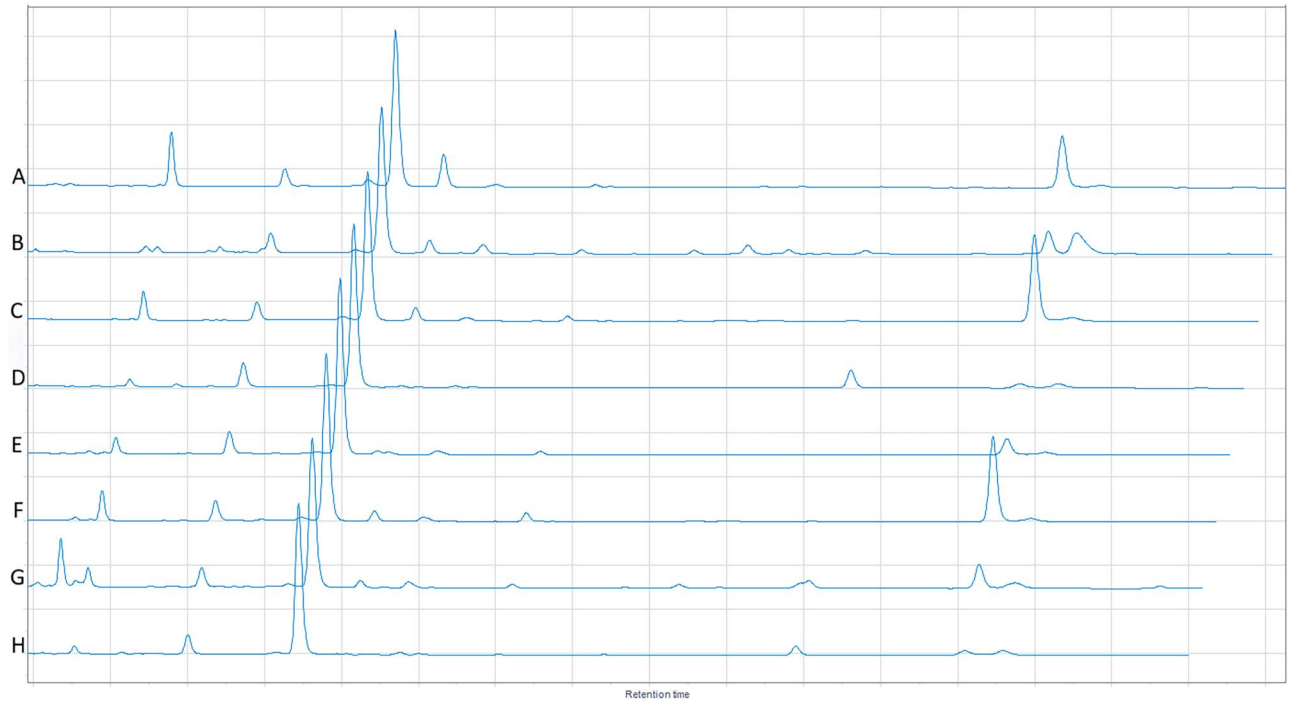


Figure 2. Stacked overlay of all eight 7-HMG products analyzed in this study. The overlaid chromatograms are identified as follows: Brand 1 Package 1 (A), Brand 1 Package 2 (B), Brand 1 Package 3 (C), Brand 2 Package 1 (D), Brand 2 Package 2 (E), Brand 3 (F), Brand 4 (G), and Brand 5 (H). A 1% time offset between the chromatograms has been used for better visualization of peaks.

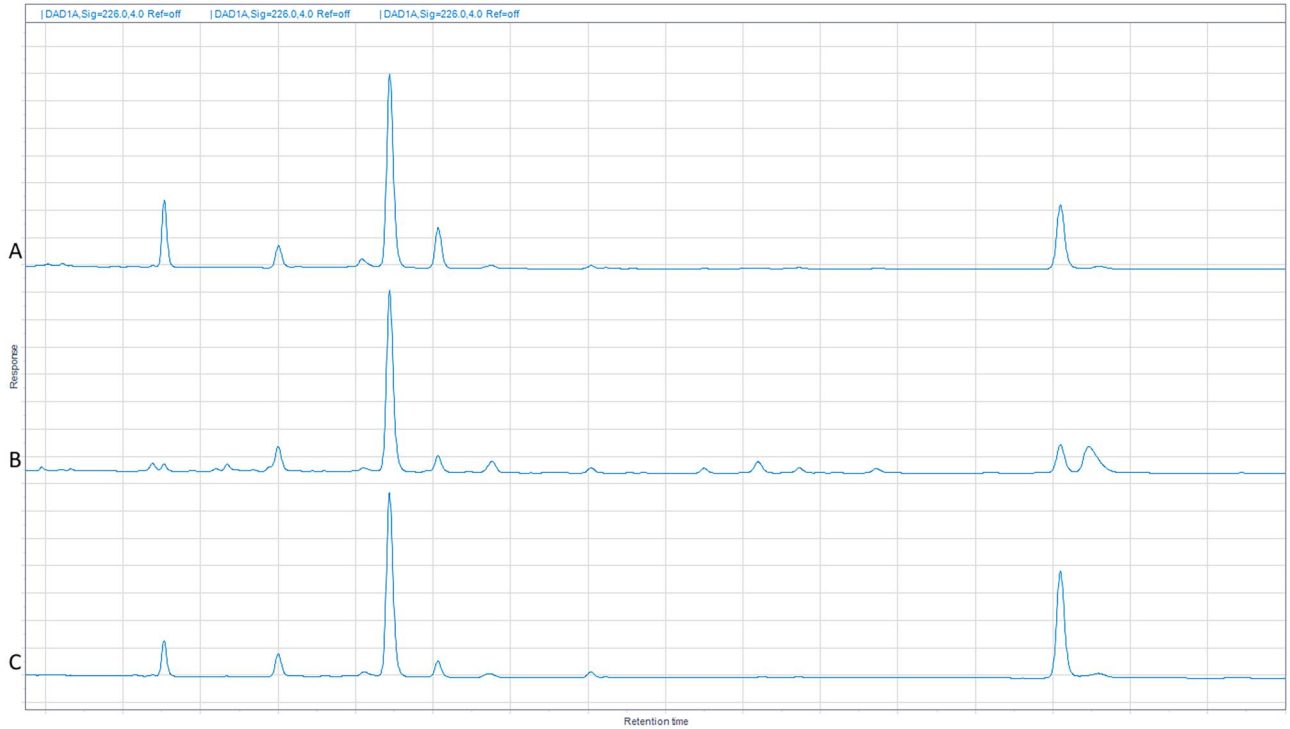


Figure 3. Chromatograms obtained from the analysis of three different packages of the same 7-HMG product. The overlaid chromatograms are identified as follows: Brand 1 Package 1 (A), Brand 1 Package 2 (B), and Brand 1 Package 3 (C). All packages were purchased from the same vendor at the same time.

chemical profiles no longer resemble kratom leaves and cannot be considered equivalent chemically, functionally, or biologically to traditional kratom preparations.

It has been proposed that the amount of 7-HMG should be limited in products, because it is not naturally occurring in native kratom leaf and is found only at trace levels in processed post-

harvest leaf material (38). This is a reasonable approach and would serve to help consumers differentiate kratom products from the synthetic unapproved drugs that have been masquerading as kratom. Another consideration that must not be ignored is that as long as 7-HMG products marketed as "kratom" are not removed from commerce, the adverse events associated with these products may be incorrectly attributed to kratom. This could negatively impact the regulation of kratom because policymakers and regulators may not differentiate between traditional kratom formulations and these unapproved synthetically produced drugs. Future considerations should include approaches for further differentiating traditional kratom leaf products from synthetic products post-consumption beyond mitragynine, which is often used as a biological marker postmortem (28).

Conclusions

This study shows that marketed 7-HMG products contain very high levels of 7-HMG and lack the chemical features of kratom leaf. To achieve these levels, a chemical conversion of mitragynine to 7-HMG is required, a process which for the products tested creates additional, unidentified compounds also not native to kratom leaf. These synthetic products were also found to be inconsistent with respect to 7-HMG content and chemical profile, demonstrating an alarming lack of control in manufacturing. 7-HMG products are misrepresented as kratom products and pose a serious concern due to the lack of proper manufacturing, characterization, and safety data. The marketing of these products is unlawful, and it is in the best interest of public health that these unapproved, synthetic products are removed from the market.

CRediT Author Statement

Paula N. Brown (Conceptualization [Lead], Funding acquisition [Lead], Investigation [Equal], Methodology [Lead], Project administration [Lead], Resources [Lead], Supervision [Lead], Writing—original draft [Equal], Writing—review & editing [Lead]), Michael Chan(Conceptualization [Supporting], Formal analysis [Lead], Investigation [Equal], Project administration [Supporting], Supervision [Supporting], Writing—original draft [Equal]), Xiaohui Zhang(Data curation [Equal], Formal analysis [Supporting], Investigation [Supporting], Methodology [Equal]), and Thomas Brendler(Conceptualization [Supporting], Investigation [Supporting], Writing—review & editing [Supporting])

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Conflict of Interest

P.B. and T.B. serve as scientific experts and regulatory consultants on botanical dietary ingredients, including kratom. No clients had any contribution or input into this study. M.C. and X.Z. have nothing to disclose.

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